

Tutorial 11

Exercise 1: Modelling the knapsack problem

Model the optimization variant of knapsack as a 0-1 ILP.

Solution:

Given an instance $W, w_1, \dots, w_n, c_1, \dots, c_n$ we construct a 0-1 ILP as follows.

We have one decision variable b_j for each item signifying whether item j is put into the backpack or not. We want to maximize the value of the items in the backpack, hence we use the objective

$$\max \sum_{j=1}^n c_j \cdot b_j.$$

The only constraint is that the selected items fit in the backpack, i.e.,

$$\sum_{j=1}^n w_j \cdot b_j \leq W.$$

Finally, we require all variables to be binary:

$$b_j \in \{0, 1\} \text{ for all } 1 \leq j \leq n.$$

Exercise 2: Modelling the subset sum problem

Model the decision variant of subset sum as a 0-1 ILP.

Solution:

Given an instance $t, n_1, \dots, n_k$ we construct a 0-1 ILP as follows.

We have one decision variable b_j for each n_j signifying whether we put it in the sum or not. Note that subset sum is not an optimization problem, we are only interested in satisfying the constraint that a subset of the n_j sums up to t . Hence, we use the constant objective function

$$f(b_1, \dots, b_j) = 0.$$

Thus, every solution has the same value.

We express the only constraint, that the selected n_j sum up to t , as

$$\sum_{j=1}^k n_j \cdot b_j \leq t$$

and

$$\sum_{j=1}^k -n_j \cdot b_j \leq -t$$

which is equivalent to $\sum_{j=1}^k n_j \cdot b_j = t$.

Finally, we require all variables to be binary:

$$b_j \in \{0, 1\} \text{ for all } 1 \leq j \leq n.$$

Exercise 3: Modelling 3-coloring

Model the decision variant of 3-coloring (see Lecture 6, Slide 8) as a 0-1 ILP.

Solution:

Given an undirected graph (V, E) we construct a 0-1 ILP as follows.

For every vertex $v \in V$ we have three decision variables, R_v , G_v , and B_v signifying whether the vertex v is colored red, green, or blue.

As we are again concerned with a decision problem, we use a constant objective function mapping every variable assignment to 0 (as in Exercise 2).

Now, we express that every vertex has exactly one color:

$$R_v + G_v + B_v = 1 \text{ for all } v \in V.$$

Next, we express for each edge $(u, v) \in E$, u and v are not both red, not both green, and not both blue:

$$R_u + R_v \leq 1 \text{ for all } (u, v) \in E$$

$$G_u + G_v \leq 1 \text{ for all } (u, v) \in E$$

$$B_u + B_v \leq 1 \text{ for all } (u, v) \in E$$

Finally, we require all variables to be binary:

$$R_v, G_v, B_v \in \{0, 1\} \text{ for all } v \in V.$$

Exercise 4: ILP in NP

Show that 0-1 ILP is in NP.

Solution:

To show that the problem is in NP, we consider the decision variant in standard form with an additional threshold t . As the standard form is a maximization problem, we want to decide whether there is a solution whose objective is at least t .

Now, consider the following halting NTM:

Given input $c_1, \dots, c_n, a_{1,1}, \dots, a_{k,n}, b_1, \dots, b_k, t$:

1. Guess a bit vector of length n , and refer to the i -th bit as x_i .
2. Compute $\sum_{i=1}^n c_i \cdot x_i$. If this is strictly smaller than t , reject.
3. For $j = 1, \dots, k$ compute $\sum_{i=1}^n a_{j,i} \cdot x_i$. If this is strictly larger than b_j , reject.
4. Accept.

Note that the coefficients c_j and $a_{i,j}$ as well as the bounds b_j are all rationals, which implies that the computations in step 2 and 3 can be done in polynomial time. Thus, we have shown that 0-1 ILP is in NP.